

Department of Artificial Intelligence & Data Science

School of Computing, FAST-NUCES

Course Title: Natural Language Processing (NLP)

Class: MS-AI/DS

Instructor: Dr. Zohair Ahmed

Total Marks: 10

Due Date: Next class. No extension. Late submissions will not be accepted.

- Viva or quiz may be conducted in the next class based on the task.
- Students must prepare the concepts.
- Blind copy-paste is not acceptable.

Task 1 Code:

Warm-Up DL Programming Task 1: Dataset Discovery + Quick Stats

- **Goal:** Choose **one public deep learning dataset** (vision/text/audio/tabular) and do **dataset diagnostics** (EDA) to judge training readiness.
- **Tools:** Colab + Python (PyTorch / datasets / torchvision / torchaudio).

What students must do

1. **Pick dataset** (name + source) and state: **input x**, **target y**, **task type** (classification/regression/sequence).
2. **Load data** + create **train/val split** (if not provided).
3. Show **8–16 samples** (images grid / text examples / audio transcript + duration).
4. Compute & print stats:
 - **Dataset size** (train/val/test)
 - **Label distribution + imbalance ratio** (max/min)
 - **Input size/length** distribution
 - Vision: image shapes (H×W×C)
 - Text: word/token lengths (min/mean/median/p90/p95/max) + **approx vocab size**
 - Audio: duration (sec) stats + transcript length + approx vocab (if transcripts)
 - **Data quality checks:** missing labels/empty text/corrupt items (count)